Additional assumptions:

- The production cost of renewable units is assumed to be zero. Additionally, these units offer their forecasted capacity, meaning their offer quantities vary over time.
- For the bid price of price-elastic demands, use comparatively high values (relative to the generation cost of conventional units) to ensure that most demands are supplied. For inspiration, check the real bid price data in Nord Pool [link].
- A potential source for wind power forecast data is available at this link (you may normalize the data to fit your case study). Another potential source for the renewable power generation data is renewables.ninja.
- For transmission lines, you may assume a uniform reactance for all lines (e.g., 0.002 p.u., leading to a susceptance of 500 p.u.).

The 24-bus power system is illustrated in Figure 1. The slack bus of the system is node 13.

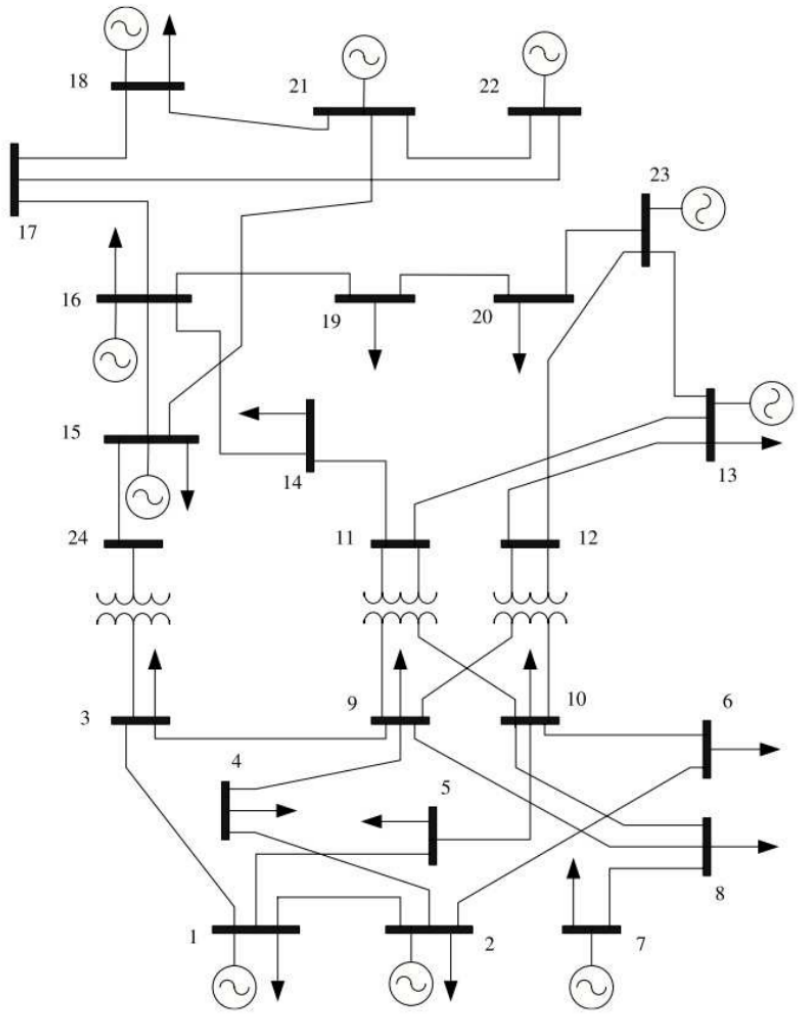


Figure 1: 24-bus power system – Single area RTS-96

Step 1: Copper-Plate, Single Hour

In Lecture 2, you learn how to develop a market-clearing optimization model for a copper-plate power system (i.e., without modeling the transmission network) in a single-hour setting.

Please determine the following market-clearing outcomes:

- The **market-clearing price** under a uniform pricing scheme. = $20.7/MWh
- The **social welfare** of the system.
- The **profit** of each producer, including both conventional units and wind farms.
- The **utility** of each demand, defined as:

$$\text{Utility} = \text{Power Consumption} \times (\text{Bid Price} - \text{Market-Clearing Price}).$$

- Additionally, please **verify** the market-clearing price using the KKT conditions.

Table F.4: Technical data for conventional generators, wind farms, demands and transmission lines

Conventional generators						1	2	3	4	5	6	7	8	9	10	11	12
Location [node]						1	2	7	13	15	15	16	18	21	22	23	23
Production cost [$/MWh]						13.32	13.32	20.7	20.93	26.11	10.52	10.52	6.02	5.47	7	10.52	10.89
Upward reserve cost [$/MW]						1.68	1.68	3.30	4.07	1.89	5.48	5.48	4.98	5.53	8.00	3.45	5.11
Downward reserve cost [$/MW]						2.32	2.32	4.67	3.93	3.11	3.52	3.52	5.02	4.97	6.00	2.52	2.89
Capacity [MW]						106.4	106.4	245	413.7	42	108.5	108.5	280	280	210	217	245
Maximum upward reserve provision capability [MW]						48	48	84	216	42	36	36	60	60	48	72	48
Maximum downward reserve provision capability [MW]						48	48	84	216	42	36	36	60	60	48	72	48
Wind farms														1	2	3	4
Location [node]														3	5	16	21
Installed capacity [MW]														500	500	300	300
Day-ahead forecast [MW]														120.54	115.52	53.34	38.16
Demands	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
Location [node]	1	2	3	4	5	6	7	8	9	10	13	14	15	16	18	19	20
Consumption [MW]	84	75	139	58	55	106	97	132	135	150	205	150	245	77	258	141	100
Curtailment cost [$/MWh]	500	500	500	500	500	500	500	500	500	500	500	500	500	500	500	500	500
Transmission lines: From node	1	1	1	2	2	3	3	4	5	6	7	8	8	9	9	10	10
To node	2	3	5	4	6	9	24	9	10	10	8	9	10	11	12	11	12
Susceptance [per-unit]	0.0146	0.2253	0.0907	0.1356	0.205	0.1271	0.084	0.111	0.094	0.0642	0.0652	0.1762	0.1762	0.084	0.084	0.084	0.084
Capacity [MW]	175	175	350	175	175	175	400	175	350	175	350	175	175	400	400	400	400
Transmission lines: From node	11	11	12	12	13	14	15	15	15	16	16	17	17	18	19	20	21
To node	13	14	13	23	23	16	16	21	24	17	19	18	22	21	20	23	22
Susceptance [per-unit]	0.0488	0.0426	0.0488	0.0985	0.0884	0.0594	0.0172	0.0249	0.0529	0.0263	0.0234	0.0143	0.1069	0.0132	0.0203	0.0112	0.0692
Capacity [MW]	500	500	500	500	250	250	500	400	500	500	500	500	500	1000	1000	1000	500

Social welfare = Consumer's valuation − Gen. cost

= cosumer surplus + prod. profit.

Consumer surplus = $\frac{1}{2}$ 2207 (500 − 20.7) = 1057815.1 $

Producer profit = $\sum_{n=1}^{i}$ DPi (MCP − MPi) = 26424.33 $

Producer cost = $\sum_{n=1}^{i}$ DPi * MPi = 19260.57 $

∴ Social Welfare = 1057815.1 + 26424.33

= 1084239.43 $

Generator dispatch and profits:

name	carrier	p_nom	marginal_cost	dispatch_mw	profit
G1	conventional	106.4	13.32	106.40	785.232
G2	conventional	106.4	13.32	106.40	785.232
G3	conventional	245.0	20.70	217.64	0.000
G4	conventional	413.7	20.93	-0.00	0.000
G5	conventional	42.0	26.11	-0.00	0.000
G6	conventional	108.5	10.52	108.50	1104.530
G7	conventional	108.5	10.52	108.50	1104.530
G8	conventional	280.0	6.02	280.00	4110.400
G9	conventional	280.0	5.47	280.00	4264.400
G10	conventional	210.0	7.00	210.00	2877.000
G11	conventional	217.0	10.52	217.00	2209.060
G12	conventional	245.0	10.89	245.00	2403.450
W1	wind	500.0	0.00	120.54	2495.178
W2	wind	500.0	0.00	115.52	2391.264
W3	wind	300.0	0.00	53.34	1104.138
W4	wind	300.0	0.00	38.16	789.912

Profits of each & every generator

Demand consumption and utilities:

name	carrier	p_nom	marginal_cost	dispatch_mw	utility
D1	demand	84.0	-500.0	84.0	40261.2
D2	demand	75.0	-500.0	75.0	35947.5
D3	demand	139.0	-500.0	139.0	66622.7
D4	demand	58.0	-500.0	58.0	27799.4
D5	demand	55.0	-500.0	55.0	26361.5
D6	demand	106.0	-500.0	106.0	50805.8
D7	demand	97.0	-500.0	97.0	46492.1
D8	demand	132.0	-500.0	132.0	63267.6
D9	demand	135.0	-500.0	135.0	64705.5
D10	demand	150.0	-500.0	150.0	71895.0
D13	demand	205.0	-500.0	205.0	98256.5
D14	demand	150.0	-500.0	150.0	71895.0
D15	demand	245.0	-500.0	245.0	117428.5
D16	demand	77.0	-500.0	77.0	36906.1
D18	demand	258.0	-500.0	258.0	123659.4
D19	demand	141.0	-500.0	141.0	67581.3
D20	demand	100.0	-500.0	100.0	47930.0

Demand consumption & utilities @ Bidding price = 500 $/MWh

@ MCP = 20.7 $/MWh

	Dispatch (MW)	Capacity (MW)	Marginal Cost ($/MWh)
W1, W2, W3, W4	327.56		0
G1	106.4	106.4	13.32
G2	106.4	"	"
G3	217.64	245	20.7
G4	0	413.7	20.93
G5	0	42	26.11
G6	108.5	108.5	10.52
G7	"	"	"
G8	280	280	6.02
G9	"	"	5.47
G10	210	210	7
G11	217	217	10.52
G12	245	245	10.89

Even though G3 has a maximum capacity of 245 MW but produces only ~217 MW.
∴ It sets the MCP at its marginal price of $20.7/MWh

$$C_g - \lambda + \bar{\mu}_g - \underline{\mu}_g = 0$$

For G3, $\bar{\mu}_g = \underline{\mu}_g = 0$ } No constraints on G3.

$$\therefore \quad 20.7 - \lambda = 0$$

$$\Rightarrow \quad \lambda = 20.7$$

Step 2: Copper-Plate, Multiple Hours

This step extends Step 1 by incorporating multiple time periods (here, 24 hours). To achieve this, introduce a new index, t, running from 1 to 24. Some input data vary across hours, such as **wind power generation, demand bid prices, and demand bid quantities**. Arbitrarily generate demand data so that demand bids are comparatively higher during peak hours. Furthermore, we now enforce *intertemporal* constraints by introducing **a large-scale energy storage unit** (e.g., pumped hydro storage) into the case study. Note that the objective function of the market-clearing problem is now the maximization of *total* social welfare over 24 hours.

Storage modeling: We use lowercase symbols for variables and Greek or uppercase symbols for parameters (input data). Let us define the following variables for storage:

- p_t^{ch} and p_t^{dis}: Charging and discharging power (MW) in hour t.
- e_t: Energy stored in the battery (MWh) in hour t.

The storage operation is subject to the following constraints:

$$0 \le p_t^{\text{ch}} \le P^{\text{ch}}, \qquad \forall t \qquad (1)$$

$$0 \le p_t^{\text{dis}} \le P^{\text{dis}}, \qquad \forall t \qquad (2)$$

$$0 \le e_t \le E, \qquad \forall t \qquad (3)$$

$$e_t = e_{t-1} + p_t^{\text{ch}} \eta^{\text{ch}} - \frac{p_t^{\text{dis}}}{\eta^{\text{dis}}}, \qquad \forall t, \qquad (4)$$

where η^{ch} and η^{dis} represent the charging and discharging efficiencies, respectively, with $\eta^{\text{ch}} < \eta^{\text{dis}} < 1$. Additionally, P^{ch}, P^{dis}, and E represent the charging capacity, discharging capacity, and energy storage capacity, respectively.

Note 1: The storage size is assumed to be on the order of other conventional units.

Note 2: For simplicity, we assume that this storage submits zero bid and offer prices to the market. Consequently, its presence does not alter the objective function formulation of the market-clearing problem. In reality, from the perspective of the storage owner, determining optimal hourly bid/offer prices for storage is a complex task that involves opportunity cost considerations.

Note 3: We consider $\eta^{\text{ch}} \neq \eta^{\text{dis}}$ to avoid simultaneous charging and discharging. Otherwise, we would need to introduce binary variables, which would prevent us from deriving prices as dual variables.

Please compute the following market-clearing outcomes:

- The **market-clearing prices** (one per hour) under a uniform pricing scheme.
- The total **social welfare** of the system over 24 hours.
- The total **profit** of each producer, including both conventional units and wind farms, over 24 hours.
- The total **profit** of the storage unit over 24 hours.

Furthermore, please analyze the following:

- Compare the hourly market-clearing prices **with and without** storage. Do you observe any interesting patterns?
- How are the hourly market-clearing prices determined in the presence of storage? Is the market-clearing price in every hour still equal to the offer price of the marginal producer? Please analyze your numerical findings.

Total S.W. without storage = 27165218.15

" " with " = 27174831.59

S.W = Consumer surplus + Gen. profits + Storage profits

	Hour	Demand_MW	Price_Without Storage	Price_With Storage
2025-01-01 00:00:00	1	1540.302	10.52	10.52
2025-01-01 01:00:00	2	1448.344	10.52	10.52
2025-01-01 02:00:00	3	1379.375	7.00	10.52
2025-01-01 03:00:00	4	1356.385	7.00	10.52
2025-01-01 04:00:00	5	1356.385	10.52	10.52
2025-01-01 05:00:00	6	1379.375	10.52	10.52
2025-01-01 06:00:00	7	1701.229	10.89	10.89
2025-01-01 07:00:00	8	1977.104	13.32	13.32
2025-01-01 08:00:00	9	2184.010	20.70	20.70
2025-01-01 09:00:00	10	2207.000	20.70	20.70
2025-01-01 10:00:00	11	2207.000	20.93	20.70
2025-01-01 11:00:00	12	2184.010	20.93	20.70
2025-01-01 12:00:00	13	2184.010	20.93	20.70
2025-01-01 13:00:00	14	2184.010	20.70	20.70
2025-01-01 14:00:00	15	2138.031	20.70	20.70
2025-01-01 15:00:00	16	2138.031	20.70	20.70
2025-01-01 16:00:00	17	2275.969	20.70	20.70
2025-01-01 17:00:00	18	2298.958	20.70	20.70
2025-01-01 18:00:00	19	2298.958	20.70	20.70
2025-01-01 19:00:00	20	2207.000	13.32	13.32
2025-01-01 20:00:00	21	2092.052	13.32	13.32
2025-01-01 21:00:00	22	1908.135	10.89	10.89
2025-01-01 22:00:00	23	1678.240	10.52	10.52
2025-01-01 23:00:00	24	1448.344	10.52	10.52

Storage considered = 300MW

Max. hours = 4

$\eta_{ch} = 0.9$

$\eta_{dis} = 0.95$

No, MCP is not always same to marginal price of a limiting gen. Instead, it depends on the gen. which was used to charge BESS during off-peak hours and when BESS was used to meet demand during peak hours.

	Profit_Without	Profit_With	Difference
G1	8710.97	8637.55	-73.42
G2	8710.97	8637.55	-73.42
G3	169.05	0.00	-169.05
G4	0.00	0.00	0.00
G5	0.00	0.00	0.00
G6	13216.38	13141.52	-74.86
G7	13216.38	13141.52	-74.86
G8	62375.60	64153.60	1778.00
G9	66071.60	67849.60	1778.00
G10	41842.50	43176.00	1333.50
G11	26432.77	26283.04	-149.73
G12	28393.05	28224.00	-169.05
W1	60062.43	61554.50	1492.07
W2	57561.07	58991.01	1429.94
W3	26578.15	27238.40	660.26
W4	19014.29	19486.64	472.35

Total profit of storage unit = 9571.33 $

Step 3: Network Constraints

In Lecture 3, you learn how to model power flow across a network and enforce power transmission network limits. Please extend your market-clearing optimization model from **Step 1 (no storage, 1-hour)** to incorporate these network constraints.

Nodal Market Prices: Once the model is extended to include the network constraints, you will derive the *nodal market-clearing prices*. These prices represent the marginal cost of supplying electricity at each node, considering both local generation and the transmission constraints. Are the nodal prices in the given hour necessarily identical?

Sensitivity Analysis: Perform a sensitivity analysis by adjusting the capacity of one or more transmission lines. This analysis will help you understand how changes in the transmission network affect the nodal prices. Discuss your results and examine how variations in transmission line capacity influence nodal price formation. When the network is congested, it would be useful to interpret how nodal prices are formed if they are not equal to the offer price of the local producer

Nodal market-clearing prices:

name	LMP_USD_per_MWh
Bus_1	19.766
Bus_2	19.859
Bus_3	16.782
Bus_4	20.143
Bus_5	20.384
Bus_6	20.747
Bus_7	20.700
Bus_8	20.700
Bus_9	20.374
Bus_10	21.026
Bus_11	23.545
Bus_12	19.598
Bus_13	20.286
Bus_14	29.276
Bus_15	11.101
Bus_16	10.520
Bus_17	10.724
Bus_18	10.819
Bus_19	12.697
Bus_20	14.585
Bus_21	10.908
Bus_22	10.835
Bus_23	15.627
Bus_24	13.296

Minimum LMP: 10.52
Maximum LMP: 29.275587691057957
LMP spread: 18.755587691057958

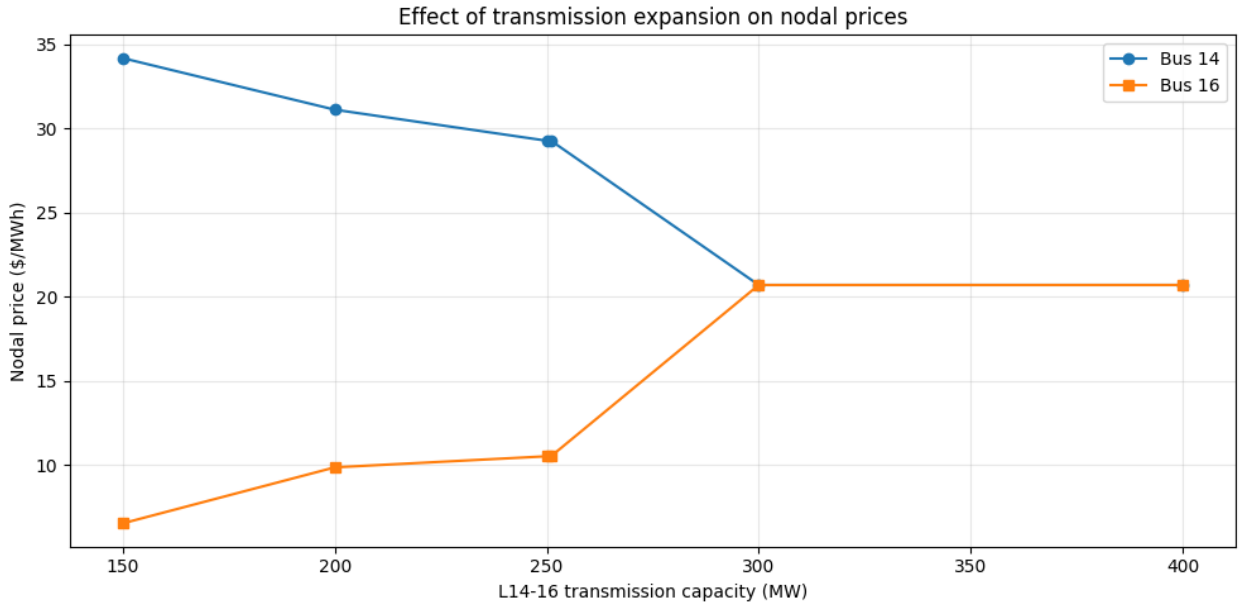
name	bus0	bus1	s_nom	Flow_MW	Loading_pct	Remaining_MW	At_limit
L14-16	Bus_14	Bus_16	250.0	-250.000	100.000	0.000	True
L15-21	Bus_15	Bus_21	400.0	-321.006	80.252	78.994	False
L13-23	Bus_13	Bus_23	250.0	-192.064	76.826	57.936	False
L16-17	Bus_16	Bus_17	500.0	-229.154	45.831	270.846	False
L3-9	Bus_3	Bus_9	175.0	68.091	38.909	106.909	False
L6-10	Bus_6	Bus_10	175.0	-63.104	36.059	111.896	False
L7-8	Bus_7	Bus_8	350.0	125.366	35.819	224.634	False
L12-23	Bus_12	Bus_23	500.0	-162.056	32.411	337.944	False
L3-24	Bus_3	Bus_24	400.0	-110.654	27.663	289.346	False
L1-2	Bus_1	Bus_2	175.0	47.845	27.340	127.155	False
L16-19	Bus_16	Bus_19	500.0	133.120	26.624	366.880	False
L17-18	Bus_17	Bus_18	500.0	-128.253	25.651	371.747	False
L2-6	Bus_2	Bus_6	175.0	42.896	24.512	132.104	False
L15-24	Bus_15	Bus_24	500.0	110.654	22.131	389.346	False
L21-22	Bus_21	Bus_22	500.0	-109.100	21.820	390.900	False
L10-12	Bus_10	Bus_12	400.0	-83.814	20.953	316.186	False
L2-4	Bus_2	Bus_4	175.0	36.349	20.771	138.651	False
L17-22	Bus_17	Bus_22	500.0	-100.900	20.180	399.100	False
L11-14	Bus_11	Bus_14	500.0	-100.000	20.000	400.000	False
L5-10	Bus_5	Bus_10	350.0	59.178	16.908	290.822	False
L10-11	Bus_10	Bus_11	400.0	-67.138	16.785	332.862	False
L15-16	Bus_15	Bus_16	500.0	73.852	14.770	426.148	False
L9-12	Bus_9	Bus_12	400.0	-57.422	14.355	342.578	False
L1-3	Bus_1	Bus_3	175.0	-24.103	13.773	150.897	False
L4-9	Bus_4	Bus_9	175.0	-21.651	12.372	153.349	False

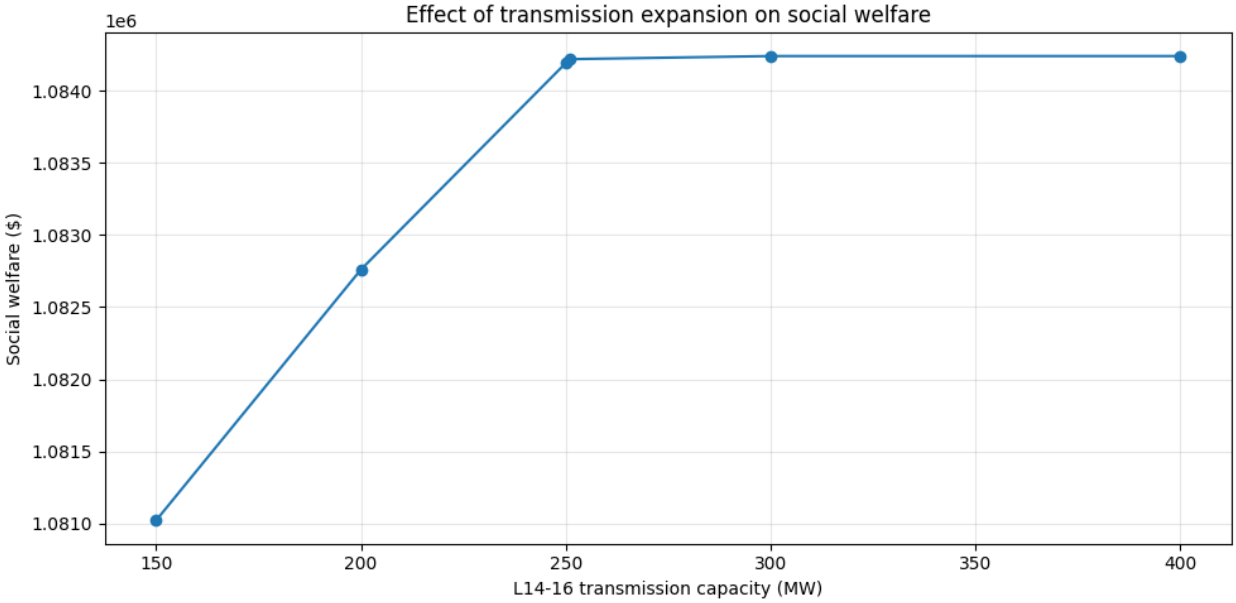
L20-23	Bus_20	Bus_23	1000.0	-107.880	10.788	892.120	False
L18-21	Bus_18	Bus_21	1000.0	-106.253	10.625	893.747	False
L9-11	Bus_9	Bus_11	400.0	-40.746	10.187	359.254	False
L8-9	Bus_8	Bus_9	175.0	-9.608	5.490	165.392	False
L12-13	Bus_12	Bus_13	500.0	20.820	4.164	479.180	False
L8-10	Bus_8	Bus_10	175.0	2.974	1.699	172.026	False
L11-13	Bus_11	Bus_13	500.0	-7.884	1.577	492.116	False
L19-20	Bus_19	Bus_20	1000.0	-7.880	0.788	992.120	False
L1-5	Bus_1	Bus_5	350.0	-1.342	0.383	348.658	False

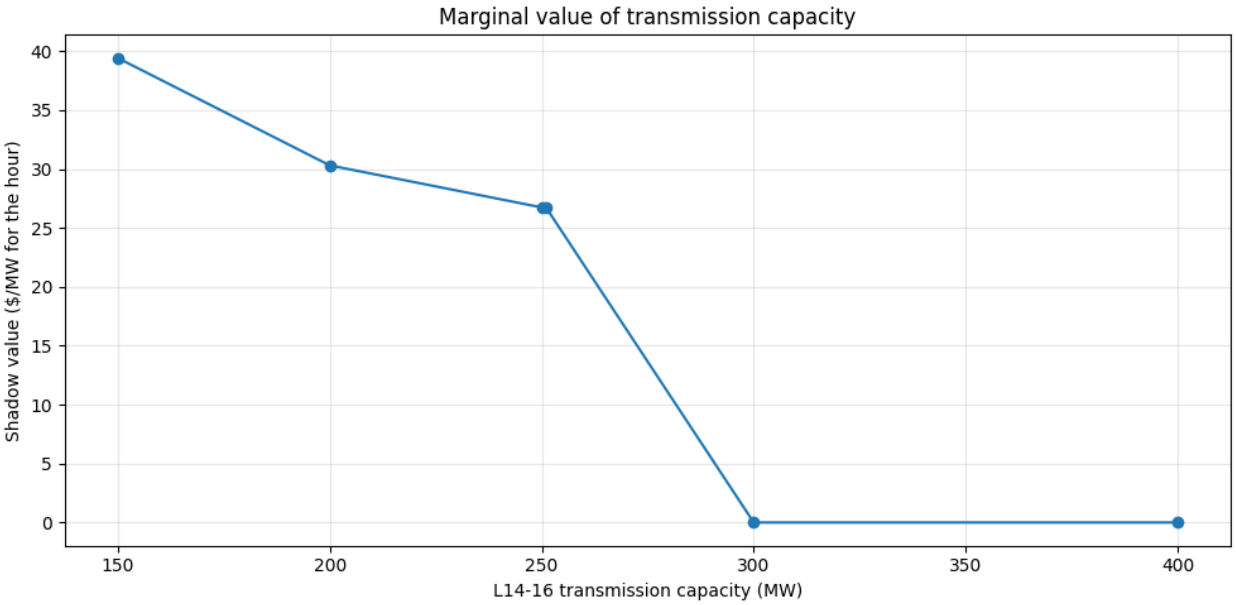
Lines at capacity: 1

Capacity_MW	Flow_MW	Loading_pct	LMP_Bus14	LMP_Bus16	LMP_Spread	Line_Shadow_Value	Production_Cost	Consumer_Value	Social_Welfare
150	-150.000	100.000	34.182	6.535	27.647	39.425	22479.137	1103500.0	1081020.863
200	-200.000	100.000	31.119	9.862	21.257	30.313	20740.233	1103500.0	1082759.767
250	-250.000	100.000	29.276	10.520	18.756	26.746	19308.688	1103500.0	1084191.312
251	-251.000	100.000	29.276	10.520	18.756	26.746	19281.941	1103500.0	1084218.059
300	-251.799	83.933	20.700	20.700	0.000	-0.000	19260.574	1103500.0	1084239.426
400	-251.799	62.950	20.700	20.700	0.000	-0.000	19260.574	1103500.0	1084239.426

As capacity of transmission increases, the social welfare increases but congestion rent decreases. This happens because a cheaper generator is able to meet the demand at another node which was not being met due to transmi-
-ssion line constraints.

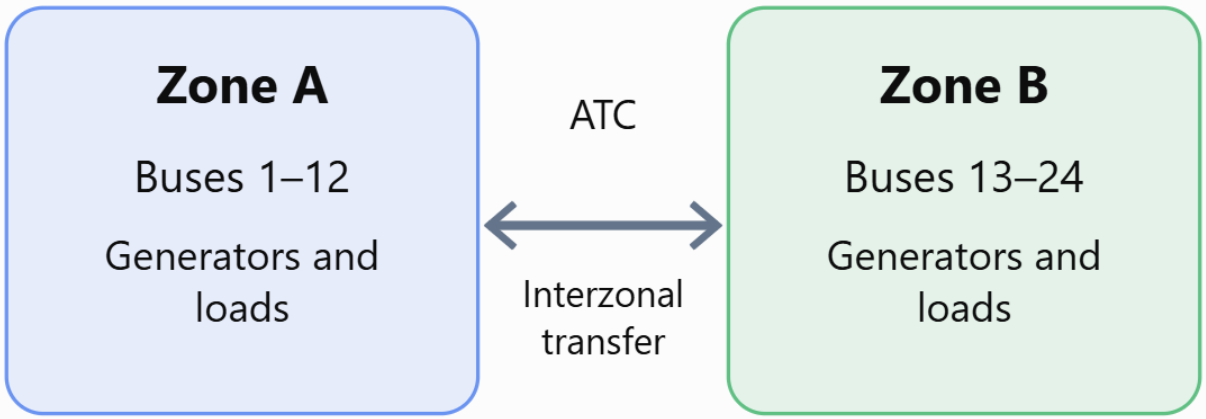

Effect of transmission expansion on nodal prices
Bus 14
Bus 16
Nodal price ($/MWh)
L14-16 transmission capacity (MW)


Effect of transmission expansion on social welfare
1e6
Social welfare ($)
L14-16 transmission capacity (MW)


Marginal value of transmission capacity
Shadow value ($/MW for the hour)
L14-16 transmission capacity (MW)

Zonal Market Prices: Consider switching to a *zonal* framework by dividing the power network into two or three zones. Within each zone, the electricity price is uniform. For simplicity, the Available Transfer Capacities (ATC) between two zones could be the total capacity of all lines between those two zones in the nodal system. How do the zonal market prices compare to the nodal prices? For different values of ATCs between the zones, derive the zonal market prices and analyze them.

Implications of Nodal vs. Zonal Frameworks: How are the profits of conventional and renewable generators affected? Compare the social welfare obtained in the nodal and zonal frameworks. Additionally, check whether the zonal market-clearing outcomes are feasible in terms of transmission line capacities within the zones. What is the magnitude of potential *ex-post re-dispatch* required to restore feasibility? You are not expected to solve a re-dispatch optimization problem, but a discussion on the magnitude of infeasibility is sufficient.




```
Zonal electricity prices:
name
Zone_A    20.7
Zone_B    20.7
Name: Hour_1, dtype: float64

Interzonal transfer: -364.500 MW
Positive = Zone A to Zone B
Production cost: $19,260.57
Social welfare: $1,084,239.43

Nodal welfare: $1,084,191.31
Zonal minus nodal welfare: $48.11

Zonal generator dispatch:
name
G1     106.40
G2     106.40
G3     217.64
G4      -0.00
G5      -0.00
G6     108.50
G7     108.50
G8     280.00
G9     280.00
G10    210.00
G11    217.00
G12    245.00
W1     120.54
W2     115.52
W3      53.34
W4      38.16
Name: Hour_1, dtype: float64
```

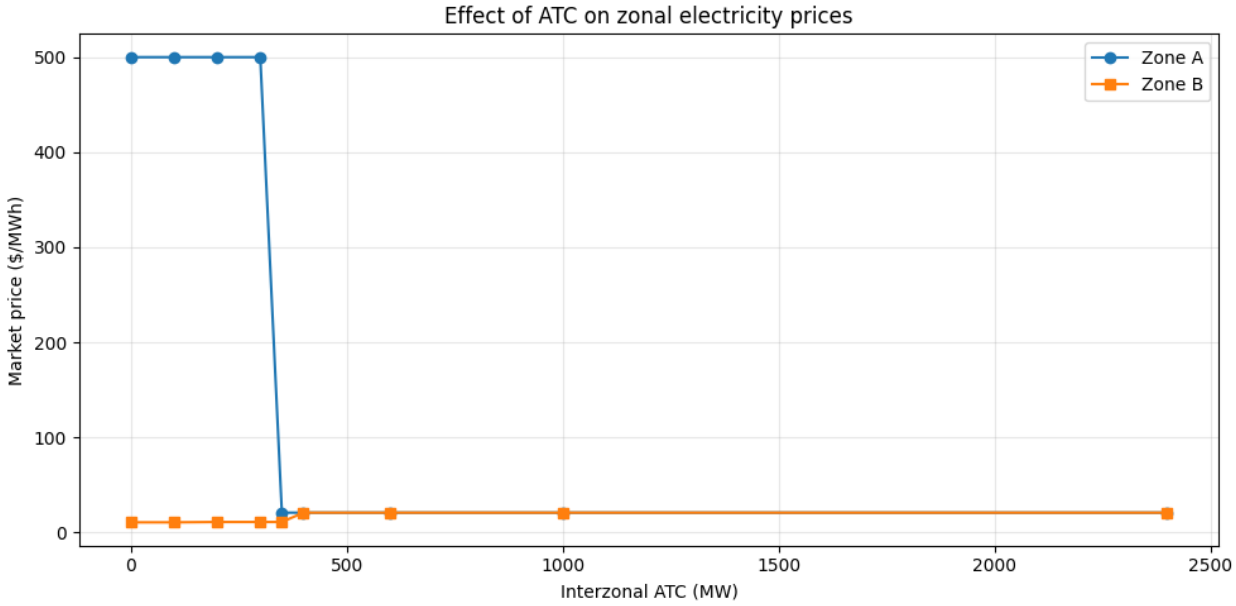
Generator	Bus	Zone	Profit_Nodal	Profit_Zonal	Difference
G1	Bus_1	Zone_A	685.822	785.232	99.410
G2	Bus_2	Zone_A	695.795	785.232	89.437
G3	Bus_7	Zone_A	0.000	0.000	0.000
G4	Bus_13	Zone_B	0.000	0.000	0.000
G5	Bus_15	Zone_B	0.000	0.000	0.000
G6	Bus_15	Zone_B	63.002	1104.530	1041.528
G7	Bus_16	Zone_B	0.000	1104.530	1104.530
G8	Bus_18	Zone_B	1343.842	4110.400	2766.558
G9	Bus_21	Zone_B	1522.596	4264.400	2741.804
G10	Bus_22	Zone_B	805.448	2877.000	2071.552
G11	Bus_23	Zone_B	1108.181	2209.060	1100.879
G12	Bus_23	Zone_B	1160.522	2403.450	1242.928
W1	Bus_3	Zone_A	2022.952	2495.178	472.226
W2	Bus_5	Zone_A	2354.802	2391.264	36.462
W3	Bus_16	Zone_B	561.137	1104.138	543.001
W4	Bus_21	Zone_B	416.243	789.912	373.669

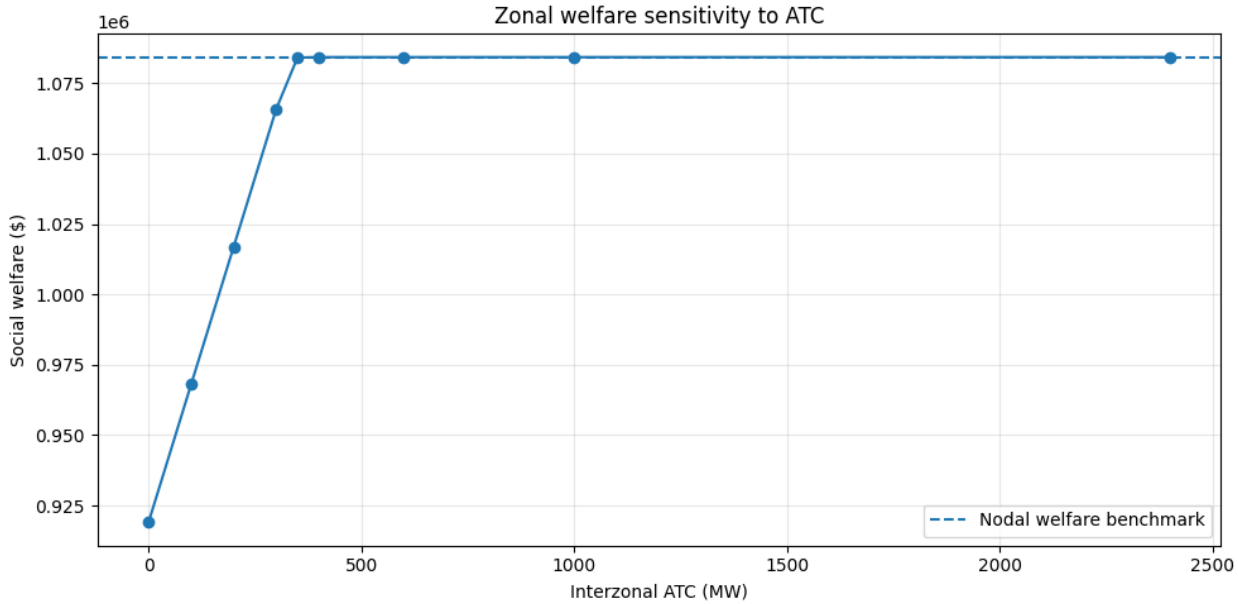
```
Total nodal generator profit: 12740.340072671546
Total zonal generator profit: 26424.326
```

We see that zonal profit is more as compared to nodal. This is because there is not much

restrictions on the capacity of the transmission line because, we add the capacities of all transmission line in a zone.

ATC_MW	Transfer_A_to_B_MW	Zone_A_Price	Zone_B_Price	Price_Spread	Cleared_Demand_MW	Production_Cost	Consumer_Value	Social_Welfare
0	-0.0	500.0	10.52	489.48	1869.86	15901.736	934930.0	919028.264
100	-100.0	500.0	10.52	489.48	1969.86	16953.736	984930.0	967976.264
200	-200.0	500.0	10.89	489.11	2069.86	18035.521	1034930.0	1016894.479
300	-300.0	500.0	10.89	489.11	2169.86	19124.521	1084930.0	1065805.479
350	-350.0	20.7	10.89	9.81	2207.00	19402.819	1103500.0	1084097.181
400	-364.5	20.7	20.70	0.00	2207.00	19260.574	1103500.0	1084239.426
600	-364.5	20.7	20.70	0.00	2207.00	19260.574	1103500.0	1084239.426
1000	-364.5	20.7	20.70	0.00	2207.00	19260.574	1103500.0	1084239.426
2400	-364.5	20.7	20.70	0.00	2207.00	19260.574	1103500.0	1084239.426





```
Total upward redispatch: 4.7263 MW
Total downward redispatch: 4.7263 MW
Net generation change: -0.0 MW
Net production-cost increase: 48.1137 USD

Generators requiring adjustments:
```

name	Bus	Zonal_MW	Nodal_MW	Marginal_Cost	Change_MW	Cost_Change_USD
G3	Bus_7	217.64	222.3663	20.70	4.7263	97.8343
G7	Bus_16	108.50	103.7737	10.52	-4.7263	-49.7206

Re-dispa-tch

G7, located @ bus 16, is relatively cheaper at $10.52/MWh, its injection contributes to the overloaded transmission path towards bus 14. The nodal dispatch reduces G7 and increases G3, which is more expensive at $20.70/MWh.

Redispatch cost

$$\Delta C = 4.7623 (20.70 - 10.52)$$

$$\Delta C = \$48.11$$

Step 4: Optimization vs. Equilibrium

In Lecture 4, you learned that there is an equivalent equilibrium problem for the market-clearing optimization problem, under which every market participant maximizes its own profit. Derive such a profit-maximization problem for the storage in **Step 2**. Then, derive its KKT conditions. There is no need for simulations or coding for this step.

Charging power limit, P^{ch}

Discharging " " , P^{dis}

Energy Capacity , E

Charging efficiency , η^{ch}

Discharging " , η^{dis}

Obj. func.

$$\max_{p^{ch}, p^{dis}, e} \sum_{t=1}^{24} \lambda_t \left(p_t^{dis} - p_t^{ch} \right) \Delta t$$

w.r.t.

$\overline{\alpha}_t^{ch}, \underline{\alpha}_t^{ch}$ ← $0 \le p_t^{ch} \le P^{ch}$ $\forall t$

$\overline{\alpha}_t^{dis}, \underline{\alpha}_t^{dis}$ ← $0 \le p_t^{dis} \le P^{dis}$ $\forall t$

$\overline{\beta}_t, \underline{\beta}_t$ ← $0 \le e_t \le E$ $\forall t$

ν_t ← $e_t = e_{t-1} + \eta^{ch} p_t^{ch} \Delta t - \frac{p_t^{dis} \Delta t}{\eta^{dis}}$ $\forall t$

$$e_{24} = e_0$$

$$e_{t-1} + \eta^{ch} p_t^{ch} \Delta t - \frac{p_t^{dis} \Delta t}{\eta_{dis}} - e_t = 0$$

Lagrangian becomes,

$$\mathcal{L} = \sum_t \lambda_t \left(p_t^{dis} - p_t^{ch} \right) \Delta t$$

$$+ \sum_t \nu_t \left(e_{t-1} + \eta^{ch} p_t^{ch} \Delta t - \frac{p_t^{dis} \Delta t}{\eta^{dis}} - e_t \right)$$

$$+ \sum_t \bar{\alpha}_t^{ch} \left(P^{ch} - p_t^{ch} \right) + \sum_t \underline{\alpha}_t^{ch} p_t^{ch}$$

$$+ \sum_t \bar{\alpha}_t^{dis} \left(P^{dis} - p_t^{dis} \right) + \sum_t \underline{\alpha}_t^{dis} p_t^{dis}$$

$$+ \sum_t \bar{\beta}_t \left(E - e_t \right) + \sum_t \underline{\beta}_t e_t$$

$$\frac{\partial \mathcal{L}}{\partial p_t^{ch}} = -\lambda_t \Delta t + \nu_t \eta^{ch} \Delta t - \bar{\alpha}_t^{ch} + \underline{\alpha}_t^{ch} = 0$$

$$\lambda_t = \frac{\nu_t \eta^{ch} \Delta t - \bar{\alpha}_t^{ch} + \underline{\alpha}_t^{ch}}{\Delta t}$$

If charging is betn. 0 & its max. limit,

$$\lambda_t = \nu_t \eta^{ch}$$

$$\frac{\partial \mathcal{L}}{\partial p_t^{dis}} = \lambda_t \Delta t - \frac{\nu_t \Delta t}{\eta^{dis}} - \overline{\alpha}_t^{dis} + \underline{\alpha}_t^{dis} = 0$$

$$\lambda_t = \frac{\nu_t}{\eta^{dis}} + \frac{\overline{\alpha}_t^{dis} + \underline{\alpha}_t^{dis}}{\Delta t}$$

$$\frac{\partial \mathcal{L}}{\partial e_t} = -\nu_t + \nu_{t+1} - \overline{\beta}_t + \underline{\beta}_t = 0$$

$$\nu_{t+1} - \nu_t = \overline{\beta}_t - \underline{\beta}_t$$

$$\nu_{t+1} = \nu_t \quad \text{... if nothing binding}$$

Step 5: Balancing Market

In Lecture 5, you learn about the balancing market. Building upon the model in **Step 1** (no storage, no transmission network, 1 hour), assume there is an unexpected failure (outage) in one of the conventional generators. Additionally, the actual production of some wind farms is lower than their day-ahead forecast (e.g., 15%), while that of the remaining wind farms is higher than their day-ahead forecast (e.g., 10%). A subset of conventional generators are potential balancing service providers. Demands are not flexible.

If possible (depending on their day-ahead schedule), each flexible conventional generator offers upward regulation service at a price equal to the day-ahead price plus 10% of its production cost. Similarly, it offers downward regulation service at a price equal to the day-ahead price minus 15% of its production cost. The load curtailment cost is €500/MWh.

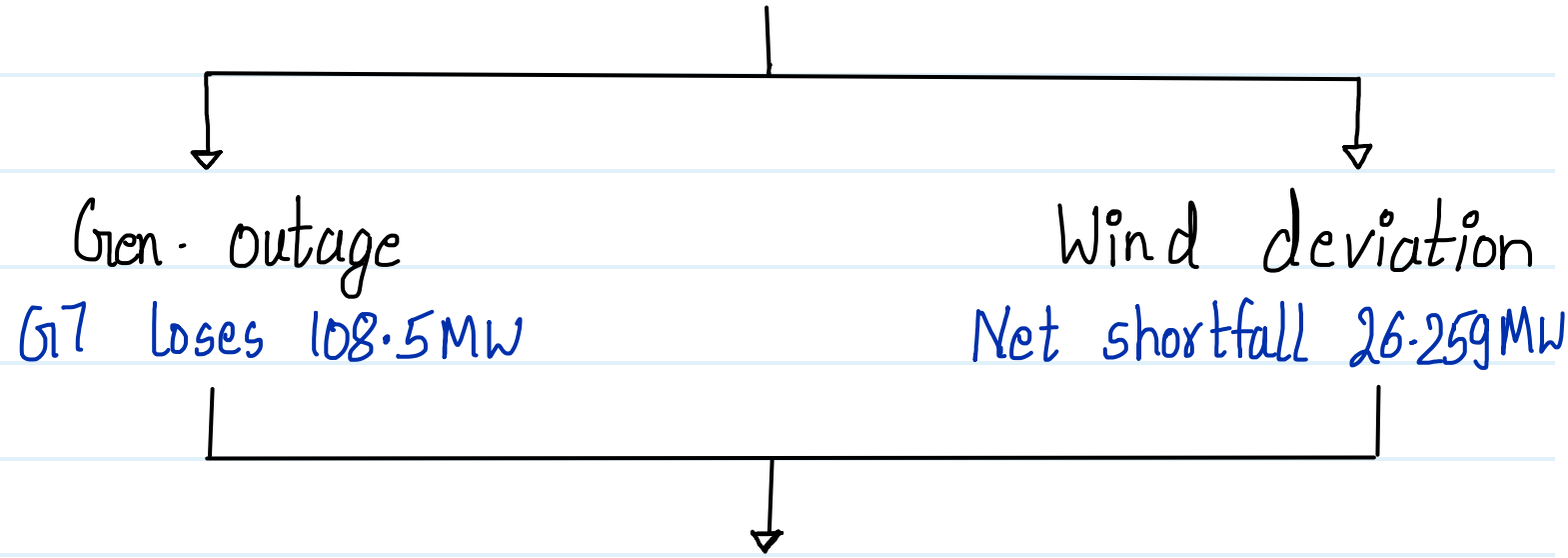
Please clear the balancing market for the given hour and derive the balancing price. Then, calculate the total profit (in both day-ahead and balancing markets) of all conventional generators and wind farms in the given hour, assuming the imbalance settlement follows:

1. One-price scheme,
2. Two-price scheme.

By comparing these profits, please analyze the implications of these two schemes for those who provide balancing services and for those who cause the imbalance.

Parameter	Selected Scenario
Conventional gen. outage	G7
W1 & W2	15% below forecast
W3 & W4	10% above "
Flexible balancing providers	G3, G4, G5, G8, G9, G11, G12
Settlement interval	1 hour

D.A. market
(Gen. = Demand)



Balancing Market Optimization

Balancing Price & Settlement

Total initial system downfall = 134.759 MW

γ_g^+ → upward regulation | L → unserved demand
γ_g^- → downward " | demand

$$\min \left[\sum_g C_g^+ \gamma_g^+ - \sum_g C_g^- \gamma_g^- + \underbrace{500L}_{\text{penalty for load shedding}} \right]$$

w.r.t.

$$\sum_g r_g^+ - \sum_g r_g^- + L = 134.759$$

$$0 \le r_g^+ \le \min\left(R_g^+, P_g^{max} - P_g^{DA}\right)$$

$$0 \le r_g^- \le \min\left(R_g^-, P_g^{DA}\right)$$

$$C_g^+ = \lambda^{DA} + 0.10 C_g$$
$$C_g^- = \lambda^{DA} - 0.15 C_g$$

name	DA_MW	Headroom_MW	Up_Max_MW	Down_Max_MW	Up_Offer	Down_Offer
G3	217.64	27.36	27.36	84.0	22.770	17.595
G4	-0.00	413.70	216.00	-0.0	22.793	17.560
G5	-0.00	42.00	42.00	-0.0	23.311	16.784
G8	280.00	0.00	0.00	60.0	21.302	19.797
G9	280.00	0.00	0.00	60.0	21.247	19.880
G11	217.00	0.00	0.00	72.0	21.752	19.122
G12	245.00	0.00	0.00	48.0	21.789	19.066

Available upward regulation: 285.3600000000001 MW

```
Balancing price: $22.793/MWh
Total upward regulation: 134.759 MW
Total downward regulation: 0.000 MW
Load shedding: -0.000 MW
Net balancing cost: $3,070.93

Upward regulation:
name
UP_G3     27.360
UP_G4    107.399
Name: Balancing_Hour, dtype: float64

Downward regulation:
Series([], Name: Balancing_Hour, dtype: float64)

Balancing constraint satisfied.
```

Unit	DA_MW	Actual_MW	Instructed_MW	Passive_Imbalance_MW	DA_Revenue	Regulation_Revenue	Profit_One_Price	Profit_Two_Price	Profit_Difference
G1	106.40	106.400	0.000	0.000	2202.480	0.000	785.232	785.232	0.000
G2	106.40	106.400	0.000	0.000	2202.480	0.000	785.232	785.232	0.000
G3	217.64	245.000	27.360	0.000	4505.148	623.616	57.264	57.264	0.000
G4	-0.00	107.399	107.399	0.000	-0.000	2447.945	200.084	200.084	0.000
G5	-0.00	-0.000	-0.000	0.000	-0.000	-0.000	0.000	0.000	0.000
G6	108.50	108.500	0.000	0.000	2245.950	0.000	1104.530	1104.530	0.000
G7	108.50	0.000	0.000	-108.500	2245.950	0.000	-227.091	-227.091	0.000
G8	280.00	280.000	0.000	0.000	5796.000	0.000	4110.400	4110.400	0.000
G9	280.00	280.000	0.000	0.000	5796.000	0.000	4264.400	4264.400	0.000
G10	210.00	210.000	0.000	0.000	4347.000	0.000	2877.000	2877.000	0.000
G11	217.00	217.000	0.000	0.000	4491.900	0.000	2209.060	2209.060	0.000
G12	245.00	245.000	0.000	0.000	5071.500	0.000	2403.450	2403.450	0.000
W1	120.54	102.459	0.000	-18.081	2495.178	0.000	2083.058	2083.058	0.000
W2	115.52	98.192	0.000	-17.328	2391.264	0.000	1996.307	1996.307	0.000
W3	53.34	58.674	0.000	5.334	1104.138	0.000	1225.716	1214.552	11.164
W4	38.16	41.976	0.000	3.816	789.912	0.000	876.890	868.903	7.987

```
Total generator and wind profits:
One-price: 24751.533
Two-price: 24732.382
```

```
Wind-farm settlement comparison:
```

Unit	Passive_Imbalance_MW	Profit_One_Price	Profit_Two_Price	Profit_Difference
W1	-18.081	2083.058	2083.058	0.000
W2	-17.328	1996.307	1996.307	0.000
W3	5.334	1225.716	1214.552	11.164
W4	3.816	876.890	868.903	7.987

```
Positive wind deviation: 9.150000000000006 MWh
Expected settlement difference: 19.151 USD
Observed settlement difference: 19.1509 USD

Settlement comparison validated.
```

name	Day_Ahead_MW	Actual_MW	Difference_MW
G1	106.40	106.400	0.000
G2	106.40	106.400	0.000
G3	217.64	245.000	27.360
G4	-0.00	107.399	107.399
G5	-0.00	-0.000	0.000
G6	108.50	108.500	0.000
G7	108.50	0.000	-108.500
G8	280.00	280.000	0.000
G9	280.00	280.000	0.000
G10	210.00	210.000	0.000
G11	217.00	217.000	0.000
G12	245.00	245.000	0.000
W1	120.54	102.459	-18.081
W2	115.52	98.192	-17.328
W3	53.34	58.674	5.334
W4	38.16	41.976	3.816

```
Total actual generation: 2207.0 MW
Real-time energy balance satisfied.
```

G3 supplies 27.36 MW

G4 supplies 107.399 MW

For wind generators,

Profit under one-price = $24751.33

Profit under two-price = $24732.38

Profit came from balancing price under one-price scheme and for W3 & W4, from DA price under two-price scheme

Step 6: Reserve Market

In Lecture 6, you learn about the reserve market. We again use the model in **Step 1**. Assume the TSO has determined that the upward reserve services to be procured in the reserve market is 15% of the total demand in the corresponding hour. This value is 10% for the downward reserve services. Consider the same subset of conventional generators in **Step 5** as flexible resources.

Please clear the reserve and day-ahead markets *sequentially*, following the current practice in European electricity markets, and report the market-clearing prices for reserve and electricity. How does the reserve market change prices in the day-ahead market?

Upward reserve = 15% of total demand
Downward reserve = 10% of total demand
Single hour demand = 2207 MW
⇒ Upward reserve = 331.05 MW
Downward " = 220.7 MW

$$\min \sum_g \left(C_g^{RU} r_g^{RU} + C_g^{RD} r_g^{RD} \right)$$

w.r.t ,

$$\sum_g r_g^{RU} \geq 331.05$$

$$\sum_g r_g^{RD} \geq 220.7$$

$$0 \leq r_g^{RU} \leq \bar{R}_g^{RU}$$

$$0 \leq r_g^{RD} \leq \bar{R}_g^{RD}$$

$$r_g^{RU} + r_g^{RD} \leq P_g^{max}$$

$$r_g^{RD} \leq P_g^{DA} \leq P_g^{max} - r_g^{RU}$$

Total demand: 2207.00 MW
Upward reserve requirement: 331.05 MW
Downward reserve requirement: 220.70 MW

Unit	Up_Cost	Down_Cost	Up_Max	Down_Max	P_Max
G3	3.30	4.67	84	84	245.0
G4	4.07	3.93	216	216	413.7
G5	1.89	3.11	42	42	42.0
G8	4.98	5.02	60	60	280.0
G9	5.53	4.97	60	60	280.0
G11	3.45	2.52	72	72	217.0
G12	5.11	2.89	48	48	245.0

Reserve market cleared successfully.

Unit	P_Max	RU_Award_MW	RD_Award_MW	Award_Cost
G3	245.0	84.00	0.0	277.2000
G4	413.7	133.05	100.7	937.2645
G5	42.0	42.00	-0.0	79.3800
G8	280.0	0.00	0.0	0.0000
G9	280.0	0.00	0.0	0.0000
G11	217.0	72.00	72.0	429.8400
G12	245.0	0.00	48.0	138.7200

Total upward reserve procured: 331.05 MW
Total downward reserve procured: 220.7 MW
Total reserve procurement offer cost: 1862.4045 USD

Upward reserve market-clearing price: 4.07 $/MW
Downward reserve market-clearing price: 3.93 $/MW

Reserve requirements and generator limits validated.

Day-ahead operating ranges after reserve procurement:

Unit	Min_DA_Output	Max_DA_Output	DA_Feasible_Range_MW
G3	0.0	161.00	161.00
G4	100.7	280.65	179.95
G5	-0.0	0.00	0.00
G8	0.0	280.00	280.00
G9	0.0	280.00	280.00
G11	72.0	145.00	73.00
G12	48.0	245.00	197.00

Original day-ahead price: 20.7
After reserve procurement: 20.93

Generator dispatch comparison:

name	Original_MW	After_Reserve_MW	Change_MW
G1	106.40	106.40	0.00
G2	106.40	106.40	0.00
G3	217.64	161.00	-56.64
G4	-0.00	128.64	128.64
G5	-0.00	-0.00	0.00
G6	108.50	108.50	0.00
G7	108.50	108.50	0.00
G8	280.00	280.00	0.00
G9	280.00	280.00	0.00
G10	210.00	210.00	0.00
G11	217.00	145.00	-72.00
G12	245.00	245.00	0.00
W1	120.54	120.54	0.00
W2	115.52	115.52	0.00
W3	53.34	53.34	0.00
W4	38.16	38.16	0.00

Changed generator schedules:

name	Original_MW	After_Reserve_MW	Change_MW
G3	217.64	161.00	-56.64
G4	-0.00	128.64	128.64
G11	217.00	145.00	-72.00

Total dispatch change: 0.0 MW

	Original_DA	After_Reserve	Change
Electricity_Price	20.700	2.093000e+01	0.2300
Production_Cost	19260.574	2.002312e+04	762.5472
Consumer_Value	1103500.000	1.103500e+06	0.0000
Social_Welfare	1084239.426	1.083477e+06	-762.5472

All reserve-constrained generator limits satisfied.

Original elec. price = $20.7/MWh
After reserve procurement } = $20.9/MWh

Optional Task: (*with an additional 10% points, but the total grade will not exceed 100%. You can go beyond the page limit by half a page.*) Please clear the U.S.-style market (the *joint* reserve and energy market in the day-ahead stage). Compare the market-clearing outcomes (schedules and prices) achieved with those of the European style.

Joint clearing optimization:

$$\min \left[\sum_g C_g p_g + \sum_g C_g^{RU} r_g^{RU} + \sum_g C_g^{RD} r_g^{RD} - \sum Vddd \right]$$

w.r.t.

$$p_g + r_g^{RU} \leq p_g^{max}$$

$$r_g^{RD} \leq p_g$$

JOINT MARKET CLEARED
Electricity price: +20.7
Upward reserve price: 4.07
Downward reserve price: 4.16

	Energy_MW	RU_MW	RD_MW
G3	116.94	84.00	0.0
G4	100.70	205.05	100.7
G5	0.00	42.00	-0.0
G8	280.00	0.00	0.0
G9	280.00	0.00	0.0
G11	217.00	-0.00	72.0
G12	245.00	0.00	48.0

	Sequential	Joint	Joint_Minus_Sequential
Electricity_Price	20.9300	20.7000	-0.2300
Upward_Reserve_Price	4.0700	4.0700	0.0000
Downward_Reserve_Price	3.9300	4.1600	0.2300
Energy_Production_Cost	20023.1212	19283.7350	-739.3862
Reserve_Offer_Cost	1862.4045	1907.0445	44.6400
Combined_Cost	21885.5257	21190.7795	-694.7462

Combined cost saving: 694.7462 USD
Energy balance and reserve requirements validated.

In European style, reserve procurement was optimized independently before day-ahead electricity-market clearing.

Upward reserve price → $4.07/MW

Downward reserve price → $3.93/MW

Subsequently DA market cleared @ $20.93/MWh & $20.7/MWh in the original market clearing problem.

Joint clearing increased reserve procurement offer costs by $44.64 but reduced electricity production costs by $739.39.